```
# Load necessary packages
if (!require(readxl)) install.packages("readxl")
```

```
## Loading required package: readxl
```

```
if (!require(ggplot2)) install.packages("ggplot2")
```

```
## Loading required package: ggplot2
```

```
if (!require(GGally)) install.packages("GGally")
```

```
## Loading required package: GGally
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg   ggplot2
```

```
if (!require(ggridges)) install.packages("ggridges")
```

```
## Loading required package: ggridges
```

```
if (!require(dplyr)) install.packages("dplyr")
```

```
## Loading required package: dplyr
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
if (!require(forcats)) install.packages("forcats")
```

```
## Loading required package: forcats
```

```
library(readxl)
library(ggplot2)
library(GGally)
library(ggridges)
library(dplyr)
library(forcats)
```

```
df <- read_excel("~/Desktop/Final/Bhakti_Final/P_final/train_data.xlsx")
glimpse(df)
```

```
## Rows: 29,165
## Columns: 19
## $ ID                    <dbl> 5037048, 5044630, 5079079, 5112872, 5105858, 510…
## $ Gender                <chr> "M", "F", "F", "F", "F", "F", "M", "F", "F", "F"…
## $ `Has a car`           <chr> "Y", "Y", "N", "Y", "N", "Y", "Y", "N", "N", "Y"…
## $ `Has a property`      <chr> "Y", "N", "Y", "Y", "N", "Y", "Y", "N", "N", "N"…
## $ `Children count`      <dbl> 0, 1, 2, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, …
## $ Income                <dbl> 135000, 135000, 180000, 360000, 270000, 135000, …
## $ `Employment status`   <chr> "Working", "Commercial associate", "Commercial a…
## $ `Education level`     <chr> "Secondary / secondary special", "Higher educati…
## $ `Marital status`      <chr> "Married", "Single / not married", "Married", "S…
## $ Dwelling              <chr> "With parents", "House / apartment", "House / ap…
## $ Age                   <dbl> -16271, -10130, -12821, -20929, -16207, -13251, …
## $ `Employment length`   <dbl> -3111, -1651, -5657, -2046, -515, -3839, -1617, …
## $ `Has a work phone`    <dbl> 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, …
## $ `Has a phone`         <dbl> 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, …
## $ `Has an email`        <dbl> 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ `Job title`           <chr> "Core staff", "Accountants", "Laborers", "Manage…
## $ `Family member count` <dbl> 2, 2, 4, 1, 1, 2, 2, 3, 1, 3, 2, 2, 1, 3, 1, 2, …
## $ `Account age`         <dbl> -17, -1, -38, -11, -41, -1, -16, -21, -50, -13, …
## $ `Is high risk`        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
```

```
# Load necessary packages
packages <- c("readxl", "ggplot2", "dplyr", "ggridges", "GGally", "forcats", "plotly",
"scales", "reshape2")
install_if_missing <- function(p) { if (!require(p, character.only = TRUE)) install.pack
ages(p) }
invisible(lapply(packages, install_if_missing))
```

```
## Loading required package: plotly
```

```
##
## Attaching package: 'plotly'
```

```
## The following object is masked from 'package:ggplot2':
##
##     last_plot
```

```
## The following object is masked from 'package:stats':
##
##     filter
```

```
## The following object is masked from 'package:graphics':
##
##     layout
```

```
## Loading required package: scales
```

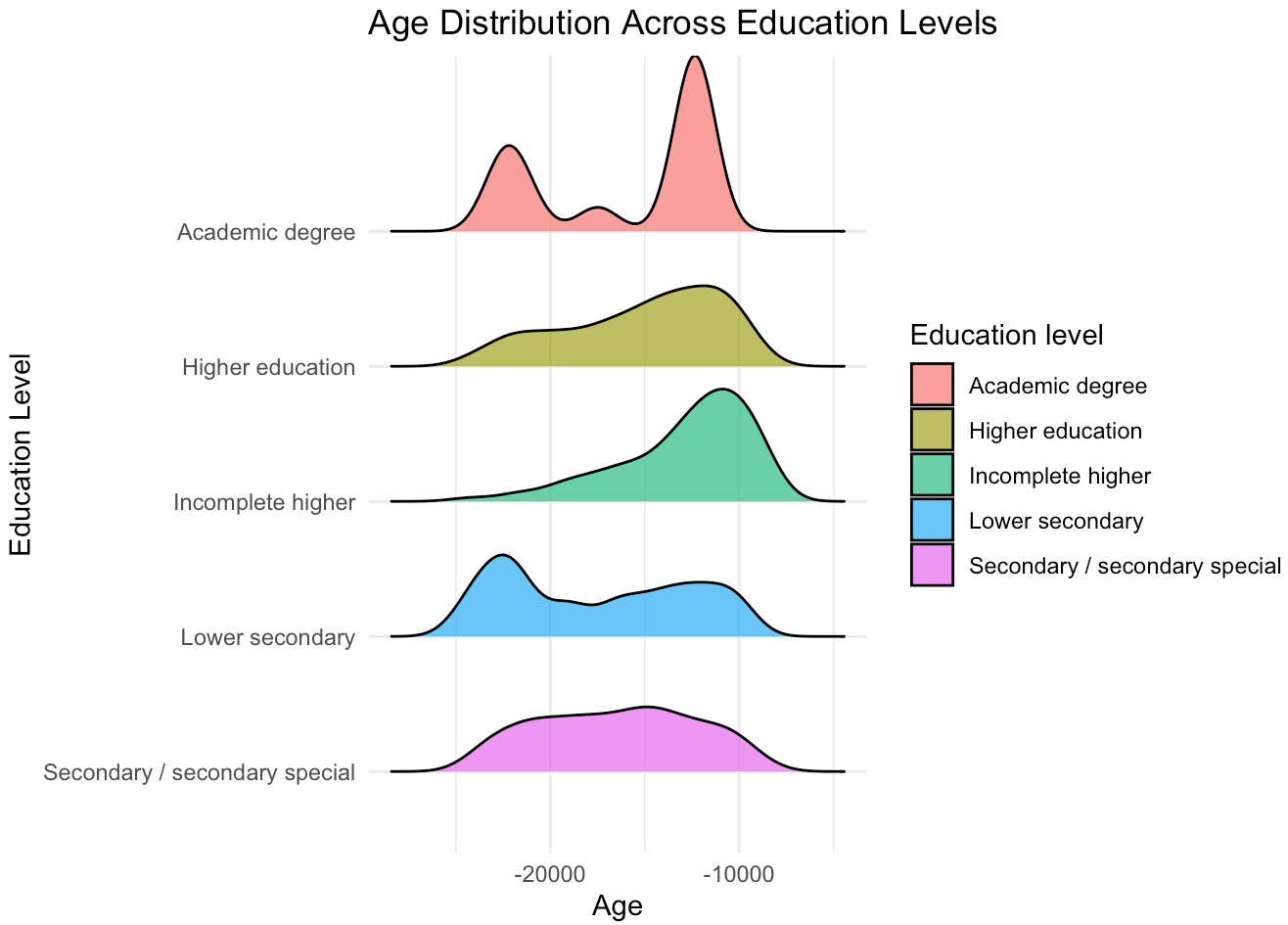
```
## Loading required package: reshape2
```

```
lapply(packages, library, character.only = TRUE)
```

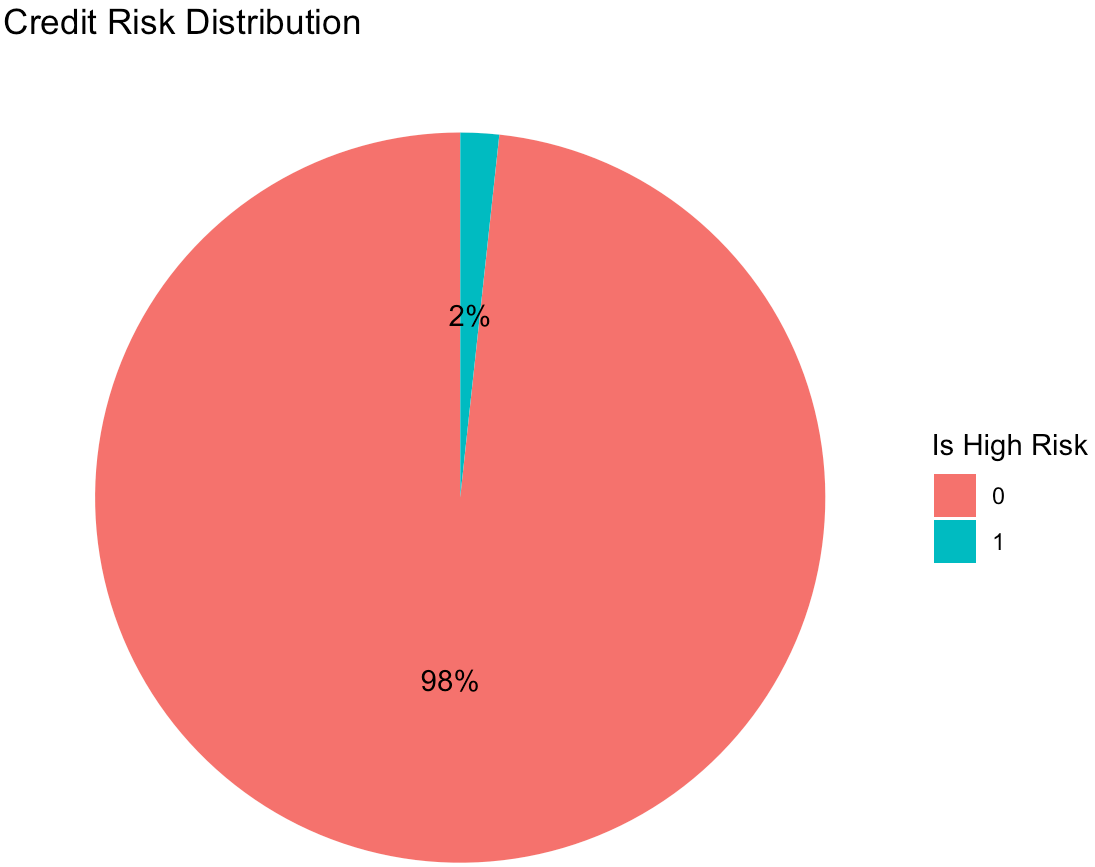
```
## [[1]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[2]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[3]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[4]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[5]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[6]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[7]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[8]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
##
## [[9]]
##  [1] "reshape2"  "scales"    "plotly"    "forcats"   "dplyr"     "ggridges"
##  [7] "GGally"    "ggplot2"   "readxl"    "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
```

```
ggplot(df, aes(x = Age, y = fct_rev(`Education level`), fill = `Education level`)) +
  geom_density_ridges(scale = 1.3, alpha = 0.7) +
  theme_minimal() +
  labs(title = "Age Distribution Across Education Levels", x = "Age", y = "Education Lev
el")
```

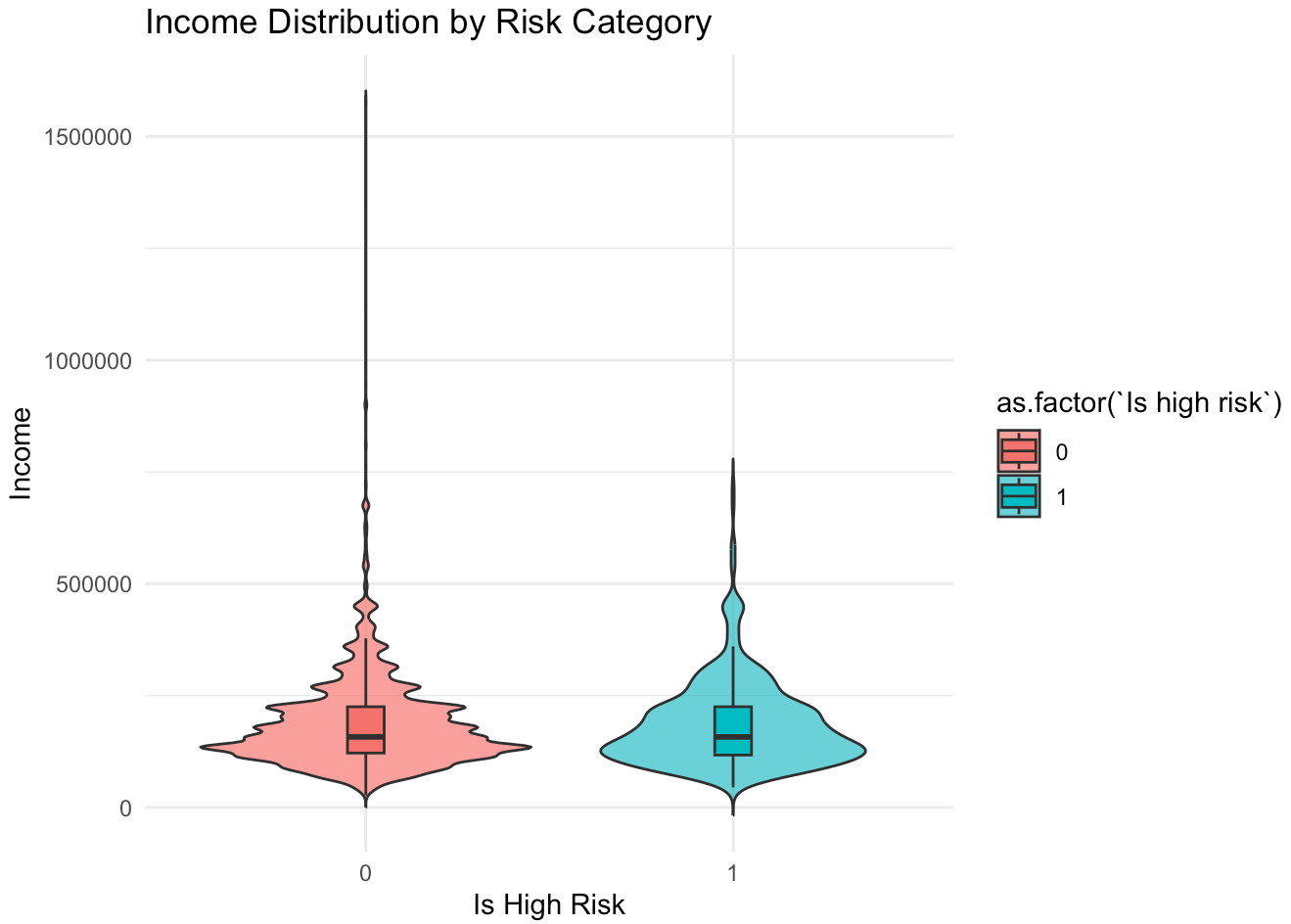
```
## Picking joint bandwidth of 1090
```



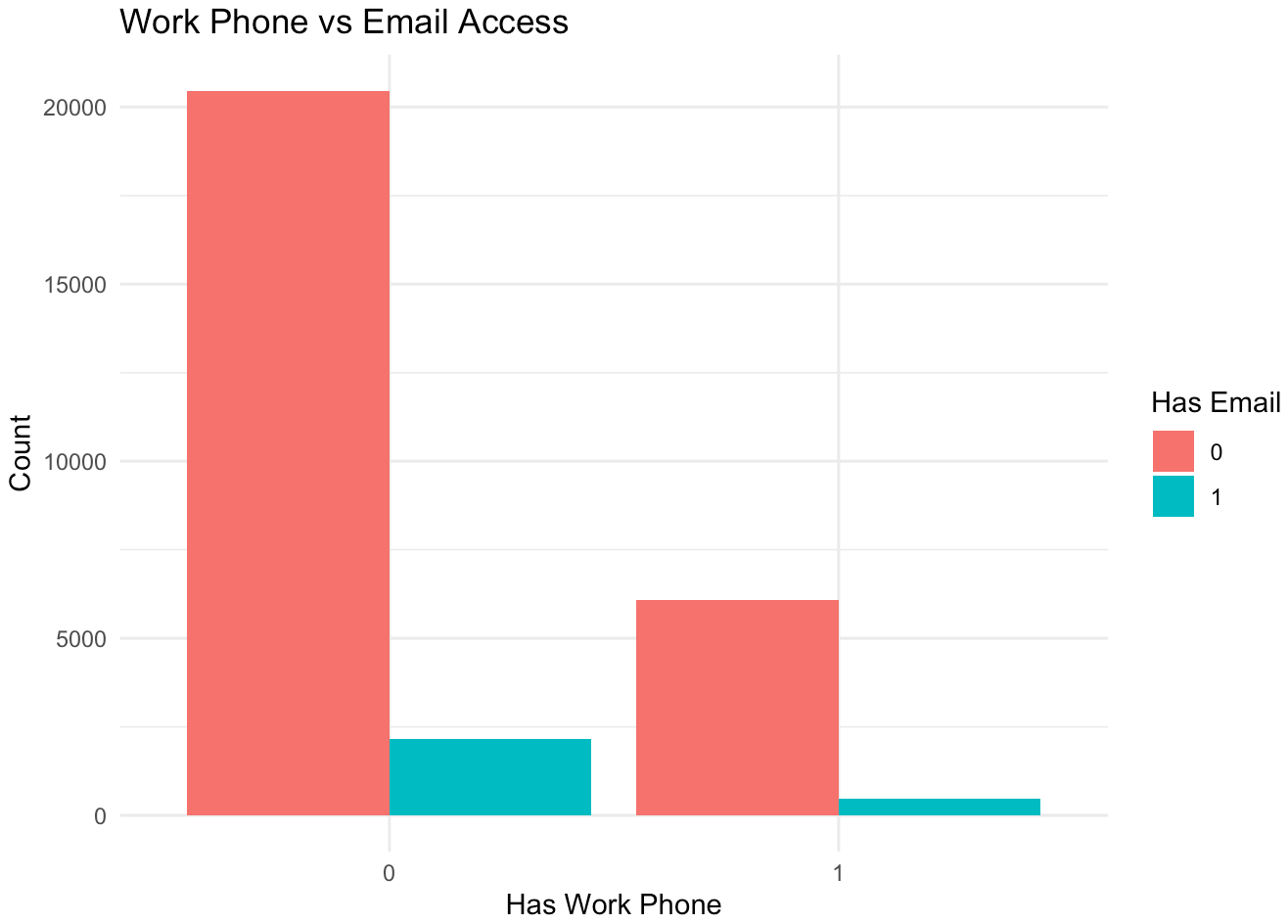

```
df %>%
  count(`Is high risk`) %>%
  mutate(percentage = n / sum(n)) %>%
  ggplot(aes(x = "", y = percentage, fill = as.factor(`Is high risk`))) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  geom_text(aes(label = scales::percent(percentage)), position = position_stack(vjust =
0.5)) +
  labs(title = "Credit Risk Distribution", fill = "Is High Risk") +
  theme_void()
```



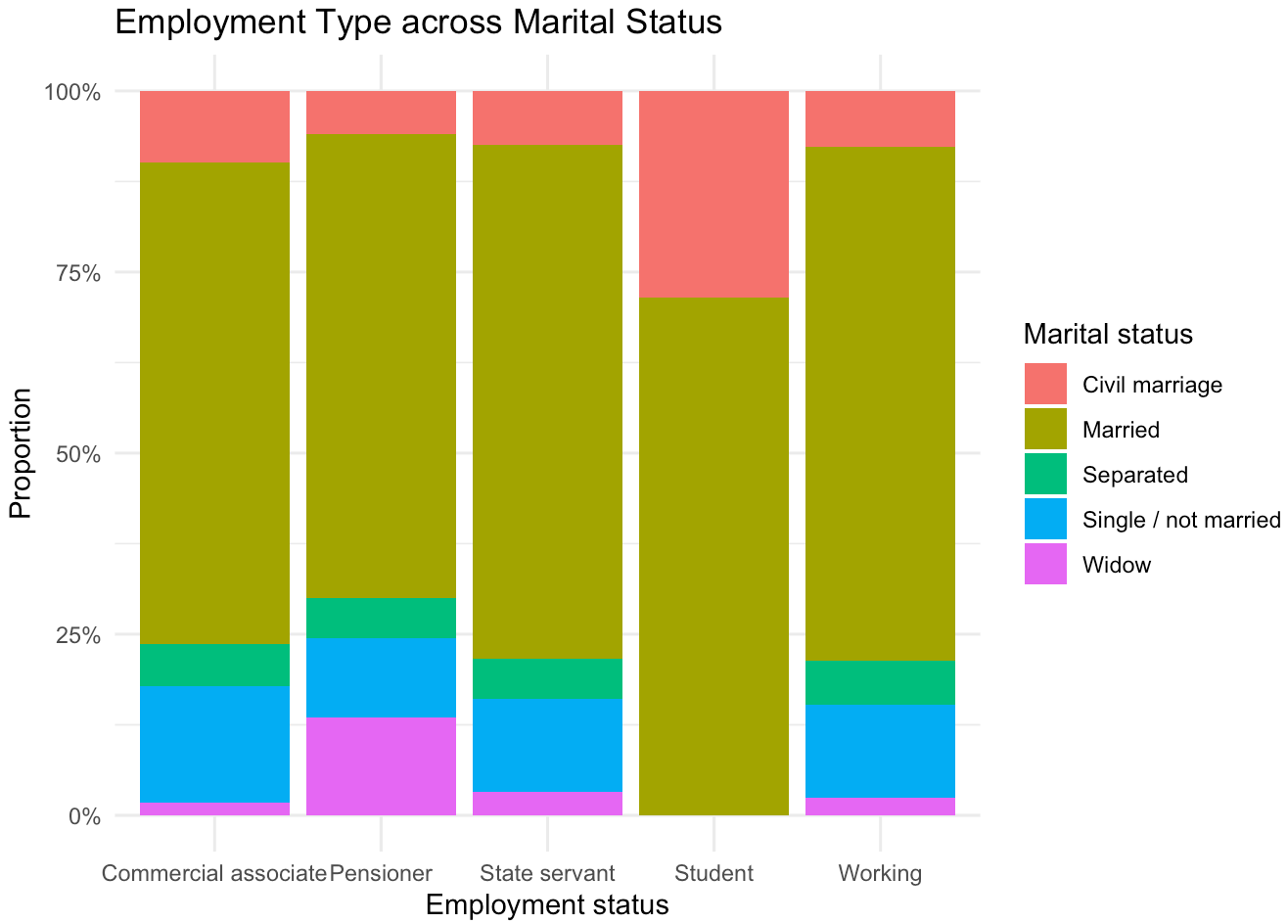

```
ggplot(df, aes(x = as.factor(`Is high risk`), y = Income, fill = as.factor(`Is high risk
`))) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_boxplot(width = 0.1, outlier.shape = NA) +
  labs(title = "Income Distribution by Risk Category", x = "Is High Risk", y = "Income")
+
  theme_minimal()
```




```
df %>%
  count(`Has a work phone`, `Has an email`) %>%
  ggplot(aes(x = as.factor(`Has a work phone`), y = n, fill = as.factor(`Has an email
`))) +
  geom_col(position = "dodge") +
  labs(title = "Work Phone vs Email Access", x = "Has Work Phone", y = "Count", fill =
"Has Email") +
  theme_minimal()
```

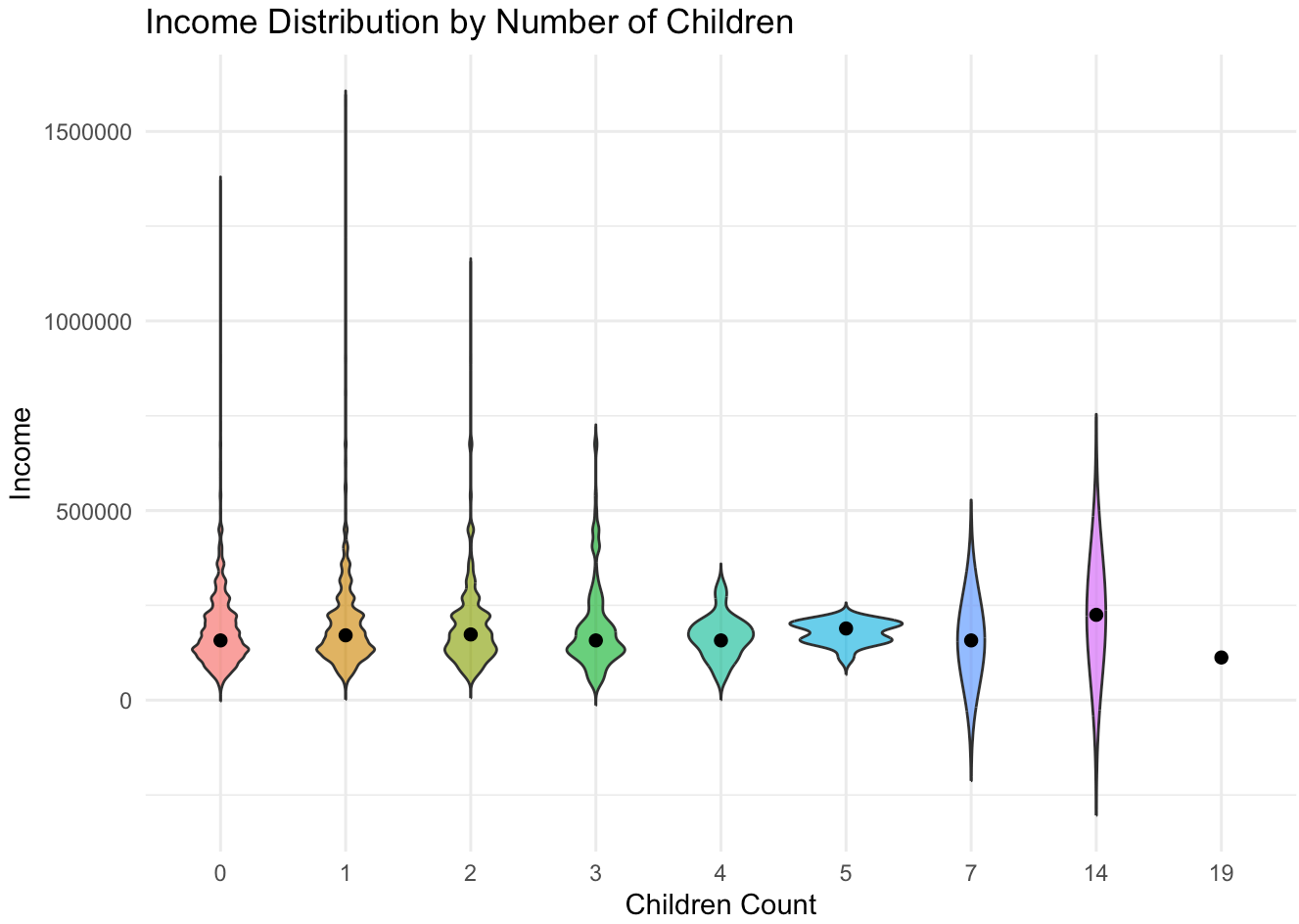



```
df %>%
  count(`Employment status`, `Marital status`) %>%
  ggplot(aes(x = `Employment status`, y = n, fill = `Marital status`)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  theme_minimal() +
  labs(title = "Employment Type across Marital Status", y = "Proportion")
```

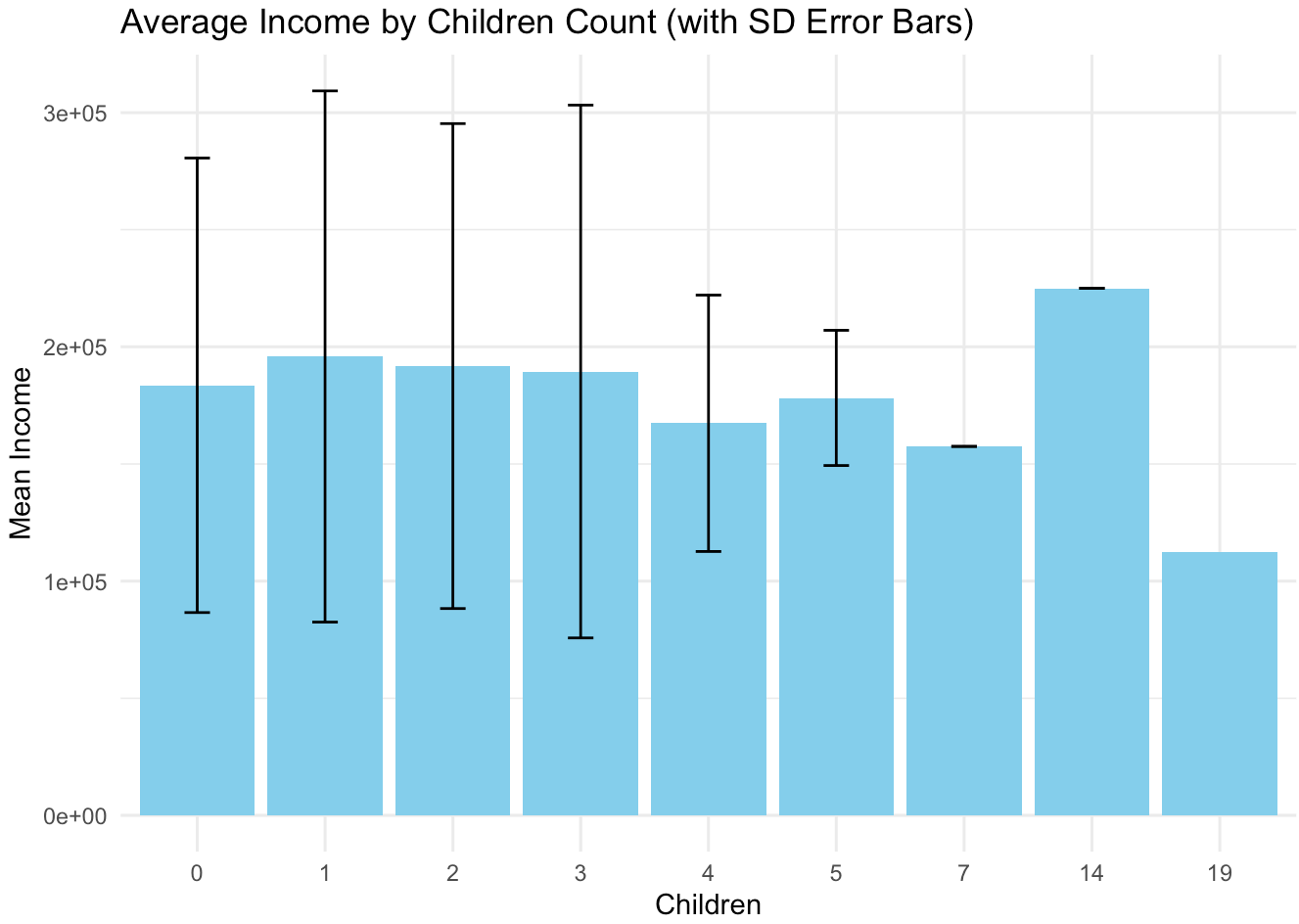



```
ggplot(df, aes(x = as.factor(`Children count`), y = Income, fill = as.factor(`Children c
ount`))) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  stat_summary(fun = "median", geom = "point", shape = 20, size = 3, color = "black") +
  theme_minimal() +
  labs(title = "Income Distribution by Number of Children", x = "Children Count", y = "I
ncome") +
  theme(legend.position = "none")
```

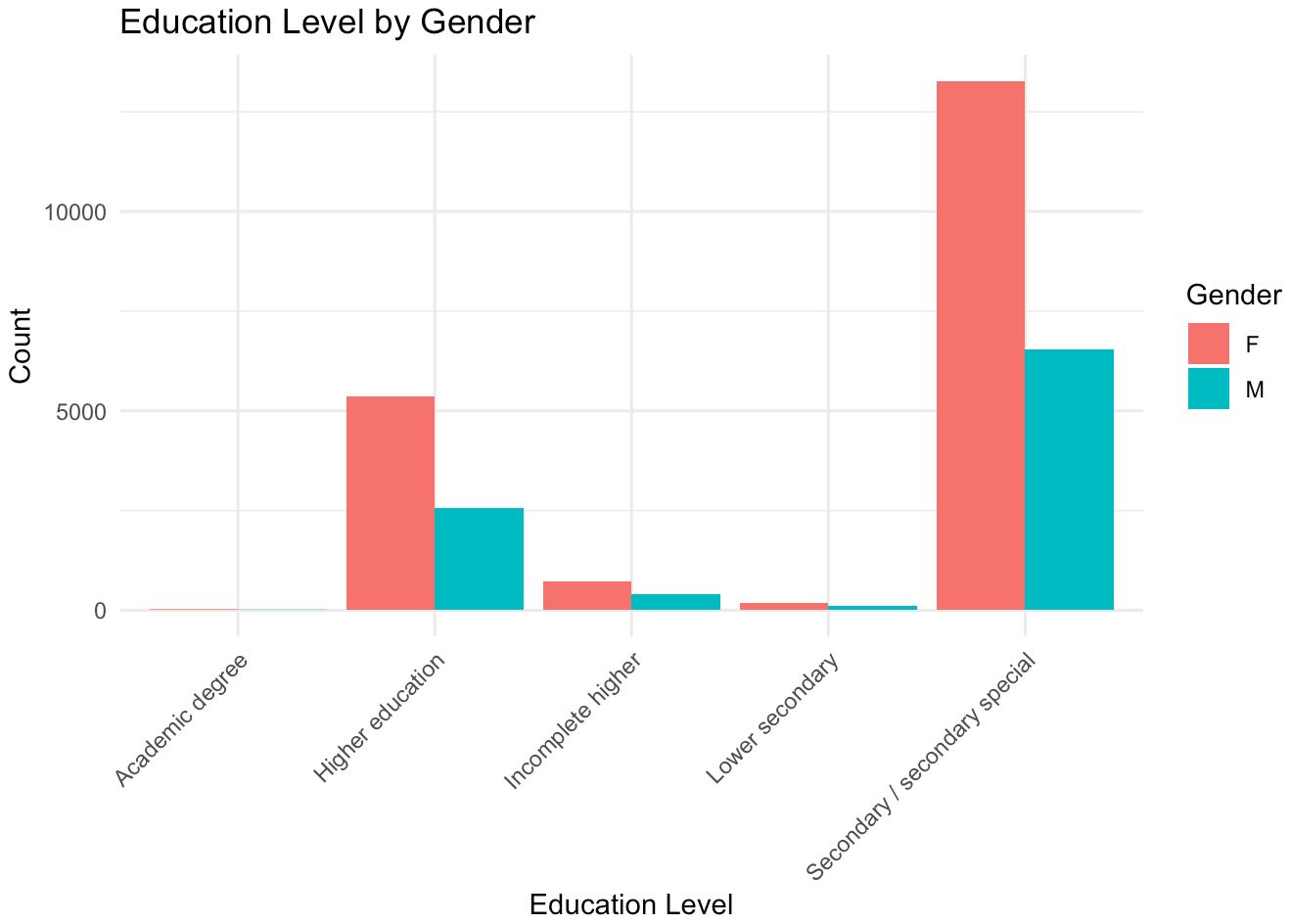
```
## Warning: Groups with fewer than two datapoints have been dropped.
## ℹ Set `drop = FALSE` to consider such groups for position adjustment purposes.
```



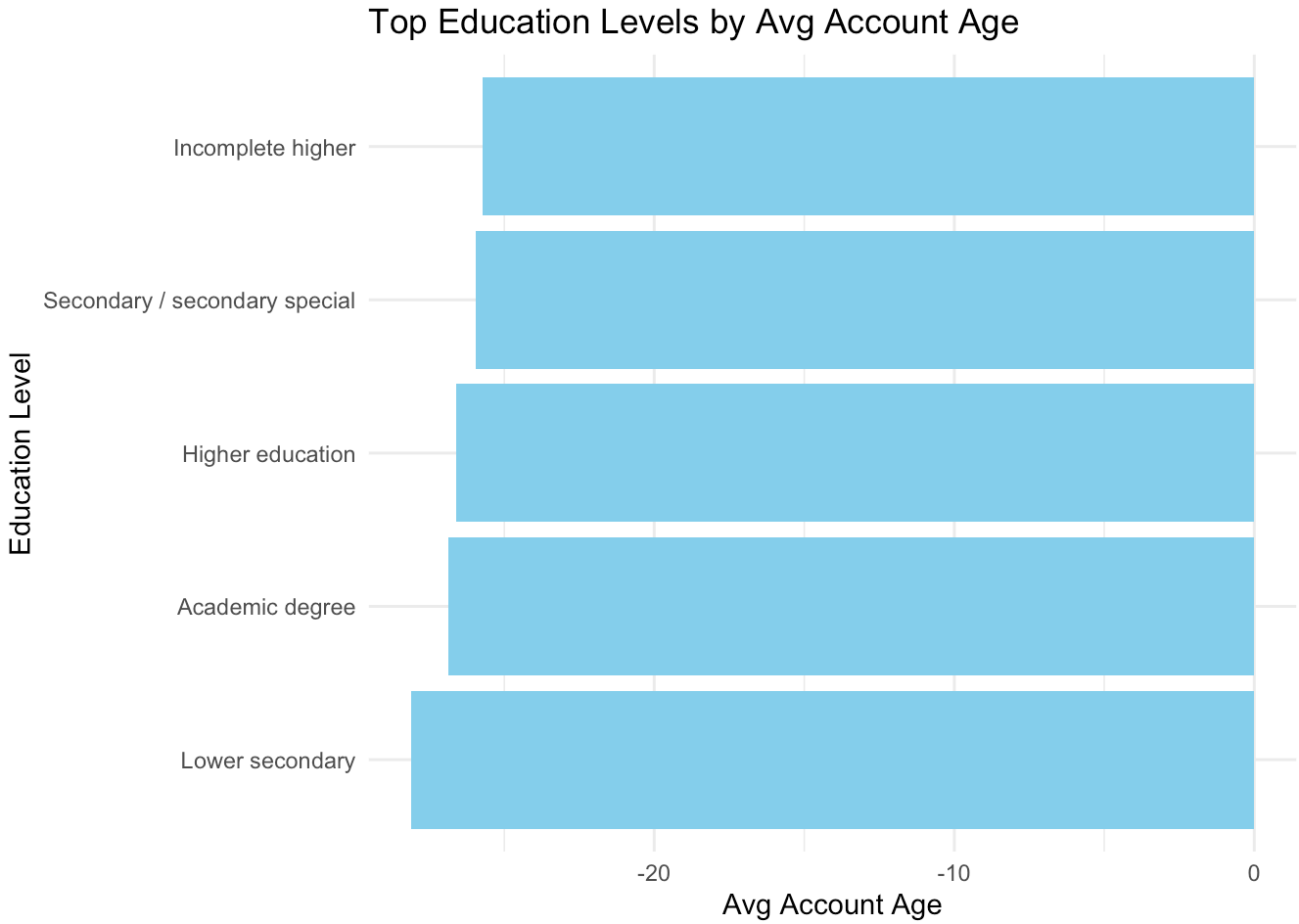

```
library(dplyr)

df %>%
  group_by(`Children count`) %>%
  summarise(mean_income = mean(Income, na.rm = TRUE),
            sd_income = sd(Income, na.rm = TRUE)) %>%
  ggplot(aes(x = as.factor(`Children count`), y = mean_income)) +
  geom_col(fill = "skyblue") +
  geom_errorbar(aes(ymin = mean_income - sd_income, ymax = mean_income + sd_income), wid
th = 0.2) +
  theme_minimal() +
  labs(title = "Average Income by Children Count (with SD Error Bars)", x = "Children",
y = "Mean Income")
```




```
ggplot(df, aes(x = `Education level`, fill = Gender)) +
  geom_bar(position = "dodge") +
  labs(title = "Education Level by Gender", x = "Education Level", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
df %>%
  group_by(`Education level`) %>%
  summarise(avg_age = mean(`Account age`, na.rm = TRUE)) %>%
  slice_max(avg_age, n = 10) %>%
  ggplot(aes(x = reorder(`Education level`, avg_age), y = avg_age)) +
  geom_col(fill = "skyblue") +
  coord_flip() +
  labs(title = "Top Education Levels by Avg Account Age", x = "Education Level", y = "Av
g Account Age") +
  theme_minimal()
```




```
library(tidyr)
```

```
##
## Attaching package: 'tidyr'
```

```
## The following object is masked from 'package:reshape2':
##
##     smiths
```

```
df %>%
  group_by(Gender, `Marital status`) %>%
  summarise(avg_income = mean(Income, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = `Marital status`, values_from = avg_income)
```

```
## # A tibble: 2 × 6
##   Gender `Civil marriage` Married Separated `Single / not married`   Widow
##   <chr>             <dbl>   <dbl>     <dbl>                  <dbl>   <dbl>
## 1 F               175667. 168824.   184820.                187873. 166506.
## 2 M               204122. 218128.   216724.                204270. 222848.
```

```r
library(tidyr)

df %>%
  group_by(`Education level`, `Employment status`) %>%
  summarise(avg_income = mean(Income, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = `Employment status`, values_from = avg_income)
```

```
## # A tibble: 5 × 6
##   `Education level`     `Commercial associate` Pensioner Working `State servant`
##   <chr>                                  <dbl>     <dbl>   <dbl>           <dbl>
## 1 Academic degree                       247500    450000  236250              NA
## 2 Higher education                      255175.   173524. 211235.         236952.
## 3 Incomplete higher                     215348.   154980  196658.         180812.
## 4 Lower secondary                       161845.   104857. 172180.         137077.
## 5 Secondary / secondar…                 197412.   144922. 171138.         182921.
## # ℹ 1 more variable: Student <dbl>
```

```r
library(tidyr)

df %>%
  group_by(`Has a car`, `Is high risk`) %>%
  summarise(avg_account_age = mean(`Account age`, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = `Is high risk`, values_from = avg_account_age,
              names_prefix = "Risk_")
```

```
## # A tibble: 2 × 3
##   `Has a car` Risk_0 Risk_1
##   <chr>        <dbl>  <dbl>
## 1 N            -25.6  -34.0
## 2 Y            -26.6  -34.2
```